Tumor Size Segmentation

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Accurate brain tumor size estimation from MRI is essential for diagnosis and treatment planning. While nn-UNet provides strong segmentation performance, it does not directly yield quantitative tumor size measurements. In this work, we extend nn-UNet by adding a method to compute tumor size and volume from segmentation outputs. By aggregating voxel-level information and incorporating a compact regression head, the framework enables direct and reliable volumetric estimation using the BraTS 2021 dataset. The proposed approach achieves high predictive performance, with an MAE of 8.88 cm³, MAPE of 15.68%, R^2 of 0.9601, and a CCC of 0.9789. These results demonstrate an accurate and efficient solution for automated tumor size and volume assessment. The implementation of the proposed method is publicly available on GitHub.

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1. INTRODUCTION

Detecting and analyzing brain tumors through magnetic resonance imaging (MRI) is a crucial aspect of medical imaging [1], as timely and precise identification of tumor regions can greatly impact diagnosis, treatment decisions, and patient outcomes [2–4]. Gliomas, which are the most common primary brain tumors, display considerable variability in shape and appearance among patients. This heterogeneity creates significant challenges for automated analysis, making accurate tumor segmentation a vital step. Clearly defining tumor boundaries helps estimate tumor size and volume, facilitates treatment planning and radiation therapy, and allows for effective monitoring of disease progression [4].

Existing methods for brain tumor analysis have largely focused on feature extraction or classification, aiming to distinguish between tumor and healthy tissue. While these approaches provide valuable insights into tumor presence, they often lack explicit spatial segmentation, which limits their interpretability and clinical utility. Without precise segmentation, automated methods cannot reliably measure tumor volume, assess growth over time, or provide detailed maps necessary for surgical or radiotherapy planning [5].

Several studies have emphasized the importance of tumor size estimation in clinical decision-making. For example, Reddy et al. [6] proposed a MATLAB-based framework for tumor size analysis, demonstrating that precise tumor extraction is fundamental for accurate volumetric assessment. Similarly, Fıçıcı et al. [7] developed an automatic detection and volume estimation method using symmetry analysis, highlighting the advantages of accurate segmentation in improving diagnostic performance. Clinical studies, including those by Cameron et al. [2] and Gray et al. [3], have further shown that tumor size at the time of detection is directly correlated with critical outcomes such as survival rate, risk of recurrence, neurological deficits, and duration of hospitalization.

Unlike prior studies that focus solely on tumor volume estimation or deep learning-based segmentation requiring large datasets [8, 9], our objective is to enhance an existing baseline framework by incorporating a lightweight, threshold-based segmentation stage. Importantly, our goal is to estimate tumor volume, keeping computational costs low. This approach positions our work between classical intensity-based segmentation methods and modern neural network pipelines, offering a practical and efficient solution for accurate tumor size estimation.

While classical and deep learning methods each have advantages, existing pipelines either lack interpretability (pure deep learning) or struggle with complex tumor morphologies (classical methods) [8]. Instead, we utilize nnU-Net as a reliable backbone to obtain tumor masks and focus exclusively on precise tumor volume computation.

In this study, we aim to address these challenges by leveraging a deep learning-based framework to estimate tumor size from MRI scans in the BraTS dataset. Our approach uses segmentation as an intermediate step to enable accurate volumetric measurements, improving the reliability and clinical relevance of automated tumor assessment. By building upon existing baseline methods, we provide a model that produces precise tumor volume estimates efficiently. Experimental results demonstrate that our method supports accurate and reliable tumor size quantification, highlighting its potential for practical clinical applications.

2. RELATED WORKS

A. Classical Brain Tumor Segmentation Methods

Early brain tumor segmentation methods relied on intensity-based and morphological operations such as thresholding, region growing, and clustering techniques (e.g., K-means and Fuzzy C-means) [10]. These approaches are computationally efficient and interpretable, making them suitable for early computer-aided diagnosis systems. However, they often struggle with heterogeneous tumor structures, low-contrast MRI regions, and intensity inhomogeneity, and their performance is highly sensitive to parameter selection and manual tuning.

To improve robustness, some studies incorporated anatomical priors or symmetry-based analysis [7]. Although these strategies enhance structural awareness, they remain sensitive to noise

and may fail when tumor morphology deviates from expected patterns.

Statistical modeling methods were later introduced to address these limitations. Pravitasari et al. [11] applied a Gaussian Mixture Model (GMM) with Expectation-Maximization for tumor segmentation and 3D reconstruction to enable volumetric estimation. This work is closely related to ours due to its focus on tumor volume computation. However, as the segmentation relies mainly on intensity distribution modeling, performance may degrade for irregular boundaries or heterogeneous textures. In contrast, our approach utilizes nnU-Net to learn spatial and contextual features, resulting in more reliable boundary delineation and stable volume estimation.

Similarly, Basthikodi et al. [5] proposed a hybrid framework combining improved K-means, Fuzzy C-means, Support Vector Machines (SVM), and Artificial Neural Networks (ANN) for tumor detection, localization, and staging. While their multi-stage pipeline achieved strong classification performance, it depends on handcrafted feature extraction, which may limit scalability and generalization. In contrast, our method employs a fully automated deep learning framework without manual feature engineering, reducing parameter sensitivity and improving robustness.

B. Deep Learning-Based Segmentation

The introduction of deep convolutional neural networks significantly improved medical image segmentation performance. U-Net and its variants have become the standard architecture for biomedical segmentation tasks [12]. Nn-UNet [8] further advances this paradigm by automatically adapting preprocessing, architecture configuration, and training strategies to dataset characteristics, achieving state-of-the-art performance on benchmarks such as BraTS [13]. Our work builds directly upon this framework due to its robustness and self-configuring capability.

Despite these advances, challenges remain in segmenting small tumor regions and low-contrast areas. Zhang et al. [14] explored supervised, transfer learning, and self-supervised strategies for detecting small metastatic brain tumors using a limited dataset of 33 MRI cases. By employing U-Net and Swin UNETR architectures, they demonstrated that deep models can be trained under data-scarce conditions. This work is particularly relevant to our study because it highlights the difficulty of small tumor segmentation an issue also encountered in deep learning systems. However, their reported Dice score for small tumors remained relatively low, indicating limitations in capturing fine grained tumor boundaries. In contrast, our work enhances segmentation refinement within the nn-UNet framework by integrating a lightweight threshold based stage to improve spatial precision without introducing complex training strategies.

In a clinically oriented study, Barhoumi et al. [15] evaluated AI-driven segmentation models for glioblastoma on T1c and FLAIR MRI sequences and compared performance with expert neuro-radiologists. Their system achieved Dice scores comparable to specialists and demonstrated low inter-user variability in volumetric measurements. This study is highly relevant to our work because it emphasizes the importance of reliable and reproducible tumor volume estimation in clinical settings. However, their research primarily evaluates a proprietary AI platform, whereas our work focuses on methodological enhancement of an open, self-configuring framework (nn-UNet) combined with classical threshold-based refinement principles to improve interpretability and boundary clarity.

C. Tumor Volume Estimation and Clinical Significance

Beyond segmentation accuracy, precise tumor volume estimation plays a critical role in treatment planning, monitoring therapeutic response, and prognostic assessment. Traditional manual measurements often approximate tumor size using geometric assumptions, leading to inter-observer variability and reduced reproducibility.

Statistical segmentation approaches such as GMM-based ROI extraction enabled early 3D tumor reconstruction and direct volume computation [11]. Modern AI-driven systems have demonstrated improved volumetric agreement compared to manual contouring [15]. These works are directly aligned with our objective of reliable tumor quantification. However, accurate volume estimation fundamentally depends on precise boundary segmentation, particularly in cases involving irregular tumor shapes, small lesions, and low-contrast MRI regions.

Our approach addresses this dependency by leveraging nnUNet's adaptive architecture for robust segmentation and integrating a lightweight threshold-based refinement stage to enhance boundary delineation. This design bridges classical intensity-based segmentation concepts [6, 7] with modern deep learning representations, improving interpretability while preserving automation and computational efficiency.

Clinical studies have demonstrated that tumor size at detection is strongly associated with survival rate, recurrence probability, neurological deficits, and hospitalization duration [2, 3]. Therefore, improving segmentation robustness and volumetric precision is not merely a technical advancement but a clinically meaningful contribution.

3. METHODOLOGY

A. Extending nn-UNet for Brain Tumor Segmentation

Our baseline model [8] is built upon nn-UNet, a self-configuring 3D U-Net framework specifically designed for medical image segmentation. nn-UNet automatically adapts its network architecture, preprocessing, and training strategies to the characteristics of each dataset, making it highly effective for brain tumor segmentation in multi-modal MRI scans. Its performance on the BraTS 2021 benchmark [13] demonstrates robust segmentation without the need for manual hyperparameter tuning. While nn-UNet provides accurate tumor delineation, our goal is to leverage this baseline framework to estimate tumor size and volume, using the segmentation output to perform precise quantitative measurements that can support clinical assessment and treatment planning.

B. NN-UNet Scalar Head for Tumor Size Estimation

To estimate tumor volume from MRI scans, we designed a *scalar regression head* on top of the standard NN-UNet backbone. The *NN-UNet Scalar Head* operates in two main steps: feature extraction via the backbone network and voxel-wise aggregation followed by a fully connected head.

Formally, given an input image $x \in \mathbb{R}^{C\times D\times H\times W}$, the backbone produces a feature map $f \in \mathbb{R}^{C_f\times D\times H\times W}$. If the backbone returns multiple outputs (e.g., a tuple of features at different resolutions), only the primary feature map is used. The spatial dimensions are then aggregated by computing the sum over all voxels and normalizing by the total number of voxels:

$$f_{\text{avg}} = \frac{\sum_{d=1}^{D}\sum_{h=1}^{H}\sum_{w=1}^{W} f_{:,d,h,w}}{D\cdot H\cdot W}$$

This results in a compact feature vector representing the global information of the input volume. Finally, this vector is passed through a small fully connected network:

$$\text{head}(f_{\text{avg}}) = \text{Linear} \circ \text{ReLU} \circ \text{Linear}(f_{\text{avg}})$$

which outputs a single scalar representing the predicted tumor volume. The hidden layer dimension is set to 128 by default, but this can be adjusted based on the dataset or model capacity.

During training, the gradients of the backbone can be optionally frozen by setting `requires_grad = False`, which allows fine-tuning only the scalar head while keeping the backbone weights fixed.

Algorithm 1. NN-UNet Scalar Head for Tumor Size Estimation

```
procedure NNUNETSCALARHEAD(backbone, n_backbone_channels, hidden)
    Store backbone
    Initialize head ← Sequential(
        Flatten(),
        Linear(n_backbone_channels, hidden),
        ReLU(inplace=True),
        Linear(hidden, 1) )

    procedure FORWARD(x)
        feat ← backbone(x)
        if feat is tuple or list then
            feat ← feat[0]
        feat_summed ← Σ_{d,h,w} feat
        num_voxels ← x_D × x_H × x_W
        feat_avg ← feat_summed / num_voxels
        output ← head(feat_avg)
        return output

procedure _SET_REQUIRES_GRAD(module, requires_grad)
    for all p ∈ module.parameters() do
        p.requires_grad ← requires_grad
```

4. EXPERIMENTAL SETUP

A. Datasets and Preprocessing

Experiments were conducted on the BraTS 2021 dataset [13], which contains multi-modal MRI scans of glioma patients, including glioblastoma and lower-grade glioma. Each patient has four MRI modalities: T1, T1ce, T2, and FLAIR, and tumor sub-regions including enhancing tumor, tumor core, and whole tumor with edema are manually annotated. All volumes were resampled to isotropic voxel spacing of $1 \times 1 \times 1$ mm^3, non-brain tissues were removed using skull-stripping, and each modality was normalized to zero mean and unit variance. Images were cropped to the brain region and padded to a fixed size suitable for 3D network input.

B. Dataset Conversion for nnU-Net

For compatibility with nnU-Net [8], the raw BraTS data was converted into the nnU-Net format, where each dataset is stored as a separate task within the `nnUNet_raw_data` folder. Each task contains training images in `imagesTr`, optional test images in `imagesTs`, and ground-truth labels in `labelsTr`. All images are 3D NIfTI files with consistent geometry across modalities, and labels contain consecutive integers starting from zero for the background. Metadata such as modality identifiers and case lists are stored in a `dataset.json` file, following the structure used by the Medical Segmentation Decathlon. For 2D data, images can be converted to pseudo-3D by adding a singleton dimension and saved as NIfTI files.

C. Tumor Volume Extraction from Segmentation Masks

In addition to preparing the data for nnU-Net, we computed quantitative tumor volumes for each patient using the ground-truth segmentation masks. The goal was to generate a dataset of tumor size measurements to serve as targets for the scalar regression head.

For each 3D segmentation mask, the physical voxel dimensions were extracted from the NIfTI header, and the voxel volume was computed as the product of the voxel spacings in the x, y, and z directions. A binary mask representing the whole tumor region was created by combining all relevant tumor sub-regions (labels 1, 2, and 4 in BraTS). The total tumor volume in cubic millimeters was then calculated by summing the tumor voxels and multiplying by the voxel volume.

This process was applied to all cases in the BraTS dataset, resulting in a CSV file containing patient identifiers and corresponding tumor volumes. This derived dataset provides direct volumetric targets that can be used to train the nn-UNet scalar regression head to predict tumor size from MRI scans. The procedure is summarized in Algorithm 2.

Algorithm 2. Compute Whole Tumor Volume from Segmentation Mask

```
procedure WHOLETUMORVOLUMEMM3(seg_path)    ▷ Returns tumor volume in mm^3
    img ← load NIfTI image from seg_path
    seg ← img.get_fdata()
    voxel_spacing ← img.header.get_zooms()[: 3]    ▷ dx, dy, dz in mm
    voxel_volume ← product of voxel_spacing
    wt_mask ← voxels with labels {1, 2, 4}
    volume_mm3 ← sum of wt_mask × voxel_volume
    return volume_mm3
```

5. CONTRIBUTIONS & RESULTS

A. Evaluation Metrics

To evaluate the performance of the proposed model on the validation set, we employed several complementary regression metrics. The *Mean Absolute Error (MAE)* measures the average absolute difference between predicted and ground-truth tumor volumes, providing a direct interpretation of the absolute estimation error in cm^3. The *Mean Absolute Percentage Error (MAPE)* expresses the prediction error as a percentage, allowing scale-independent comparison across tumors of different sizes. The *Coefficient of Determination (R^2)* quantifies the proportion of variance in the true tumor volumes that is explained by the model predictions. Finally, the *Concordance Correlation Coefficient (CCC)* evaluates both correlation and agreement between predicted and true volumes, accounting for potential bias in addition to linear association.

B. Quantitative Results

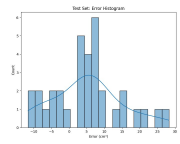
The proposed NN-UNet Scalar Head achieved strong regression performance on the validation set. Specifically, the model obtained an MAE of 8.8827 cm^3 and a MAPE of 15.68%, indicating low absolute and relative prediction errors. The R^2 score reached 0.9601, demonstrating that the model explains 96% of the variance in tumor volumes. Moreover, a CCC of 0.9789 confirms excellent agreement between predicted and ground-truth values, reflecting both high correlation and minimal systematic bias.

The visual analysis of the model's performance aligns with these numerical findings (Fig. **??**). The training trajectories for MAE, MAPE, R^2, and CCC demonstrate stable convergence over 100 epochs, with the validation Huber loss closely tracking the training loss (Fig. **??**e), indicating no significant overfitting. The scatter plot of predicted vs. actual volumes (Fig. **??**d) shows high linearity across the entire volumetric range, while the Bland-Altman analysis (Fig. **??**c) confirms that most predictions fall within the 95% limits of agreement. Finally, the error histogram (Fig. **??**a) illustrates a tight distribution centered near zero, reinforcing the model's precision.

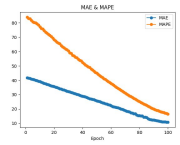
C. Training Objective

During training, we optimized the network using the Huber loss function with $\delta = 1.0$. The Huber loss combines the advantages of Mean Squared Error (MSE) and Mean Absolute Error (MAE). For small prediction errors, it behaves quadratically (similar to MSE), encouraging precise fitting. For larger errors, it transitions to a linear penalty (similar to MAE), making it more robust to outliers.

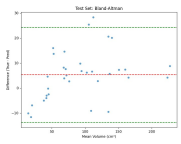
This balanced behavior is particularly suitable for medical tumor volume regression, where extreme tumor sizes may otherwise dominate the optimization process and destabilize training.



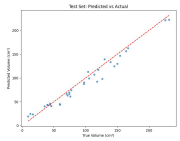
(a) Bland-Altman agreement plot.



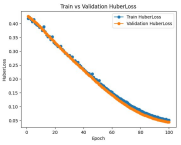
(b) MAE and MAPE training curves.



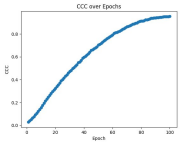
(c) Bland-Altman agreement plot



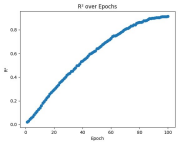
(d) Predicted vs. actual volume.



(e) Train vs. Val Huber Loss.



(f) CCC evolution over epochs.



(g) R^2 variance explanation.

Fig. 1. Qualitative results of tumor segmentation using nnU-Net.

6. CONCLUSION & FUTURE WORK

In this project, we extended the nn-UNet framework to improve brain tumor segmentation. We added a head so the model can directly predict tumor volumes while still keeping the strong segmentation performance of nn-UNet on multi-modal MRI data. This extra module adds very little computational overhead, so the model stays efficient and practical. The results show that accurate tumor segmentation can be used for quantitative tumor burden assessment without using overly complex or heavy architectures.

Our experiments were done with limited hardware. Because of restrictions on GPU memory and CPU, we couldn't do extensive hyperparameter tuning or large-scale fine-tuning. So, the results don't represent the fully optimized potential of the model. With more powerful hardware, we could improve performance through longer training, more epochs, and careful tuning.

Even with these limitations, the approach shows clear practical value. Its lightweight design makes it usable even where high-end hardware isn't available. Being able to get both segmentation maps and tumor volume estimates in one framework could make it useful in clinical settings.

For future work, we want to train the model on the full dataset to get better results. We also plan to try training for more epochs, like 5 or more, once we don't have computational limitations. Beyond that, we want to explore multi-task learning to predict both segmentation and sub-region volumes at the same time, study longitudinal MRI scans to track tumor progression, and test the model on multi-center datasets to see how well it generalizes. Eventually, we hope to integrate this into clinical decision-support systems.

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